CS383 · SOFTWARE ENGINEERING

Midterm Model 1 Solutions

Undated · 20 marks · 4 pages

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01 Solution Q1: MCQ

#	Answer	Source
1	**Security/trust vulnerabilities and internet dependence**	The SaaS slide, Drawbacks column. Lecture 2.
2	**Application type, customer requirements, team background, and tools**	The Application types slide. Lecture 1.
3	**Inflexible to changing requirements and late problem discovery**	The waterfall model slide, Weaknesses. Lecture 3.
4	**Prototyping & incremental delivery**	The Coping with change slide. Lecture 3.

Each item is taken from one specific slide, and the correct option is the slide's own wording. Note that each item is worth half a mark.

02 Solution Q2: User / System and FR / NFR

Requirement	User vs. System	FR vs. NFR
"The system should be easy to use for first-year students."	**User**	**NFR**
"The application shall allow password reset using a verified email."	**System**	**FR**
"The report generation shall be fast."	**User** *	**NFR**
"The interface will display student GPA to two decimal places."	**System**	**FR**

- User = a general statement in natural language. System = a specific, detailed statement fit for developers.
- FR = a service the system provides (allow, display). NFR = a quality attribute (easy to use, fast).
- The word shall does not make a statement functional. "shall be fast" is a quality attribute, so it is an NFR.

* This cell was blank on the solved paper. I chose User because the statement is general and has no measurable number.

03 Solution Q3: sprint calculations

1) Sprint Velocity $= \frac{22 + 28 + 26 + 24}{4} = \frac{100}{4} = 25$

2) Focus Factor $= \frac{25}{6 \times 8} = \frac{25}{48} = 0.52$

3) Working days = 3 weeks × 5 days = 15 days

Capacity = 0.52 × (6 × 8 × 15) = 0.52 × 720 ≈ 375 hours

4) 30 > 25 → Prudent

- The focus factor here is `0.52`, which is below the optimal range (0.6 to 0.8) according to the solved paper.
- In part 3 we drop two days from each week (the weekend), so a week has 5 working days.
- In part 4: the backlog (30) is slightly larger than the velocity (25), so the plan is Prudent: possible with caution, and it is better to reduce the backlog.

Case	Verdict
Backlog ≈ Velocity	**Feasible**
Backlog > Velocity (slightly)	**Prudent**
Backlog >> Velocity	**Not feasible**

04 Solution Q4: process model

Project	Answer	Evidence in the text
1 · Smart-City Sensor Hub	**Component-Based**	"Must integrate COTS gateways and vendor SDKs": the system is built from ready-made components.
2 · Banking Regulatory Reporting	**Plan-Driven**	Limited stakeholders, low volatility, 30 people, documentation and sign-offs.
3 · Campus Social-Learning MVP	**Agile**	Daily feedback, high volatility, 6 co-located developers, a beta in 4 to 6 weeks.
4 · Medical Device Companion App	**Plan-Driven**	Compliance with IEC 62304 and ISO 14971 needs full life-cycle documentation.

How to decide

Look for three signals. COTS or a ready SDK as the main constraint points to Component-Based. Daily feedback, frequent change and a small team point to Agile. Regulation, documentation, sign-offs and a large team point to Plan-Driven. Project 4 has an SDK, but compliance dominates, so the answer is Plan-Driven.

05 Solution Q5: Product / Organization / External

#	Answer	Reason
1	**Product**	The performance of the product itself (response time).
2	**Organization**	A constraint from the company: a set language and platform.
3	**External**	A law from an outside body (HIPAA).
4	**Product**	The product's run-time behaviour.
5	**Product**	It concerns the product itself. Neither the company nor an outside body imposed it.
6	**External**	An external standard (ISO/IEC) for interoperating with other systems.

The rule: Product = a property of the product itself. Organization = a policy or procedure of the customer's or developer's company. External = a law, a standard or a body from outside.